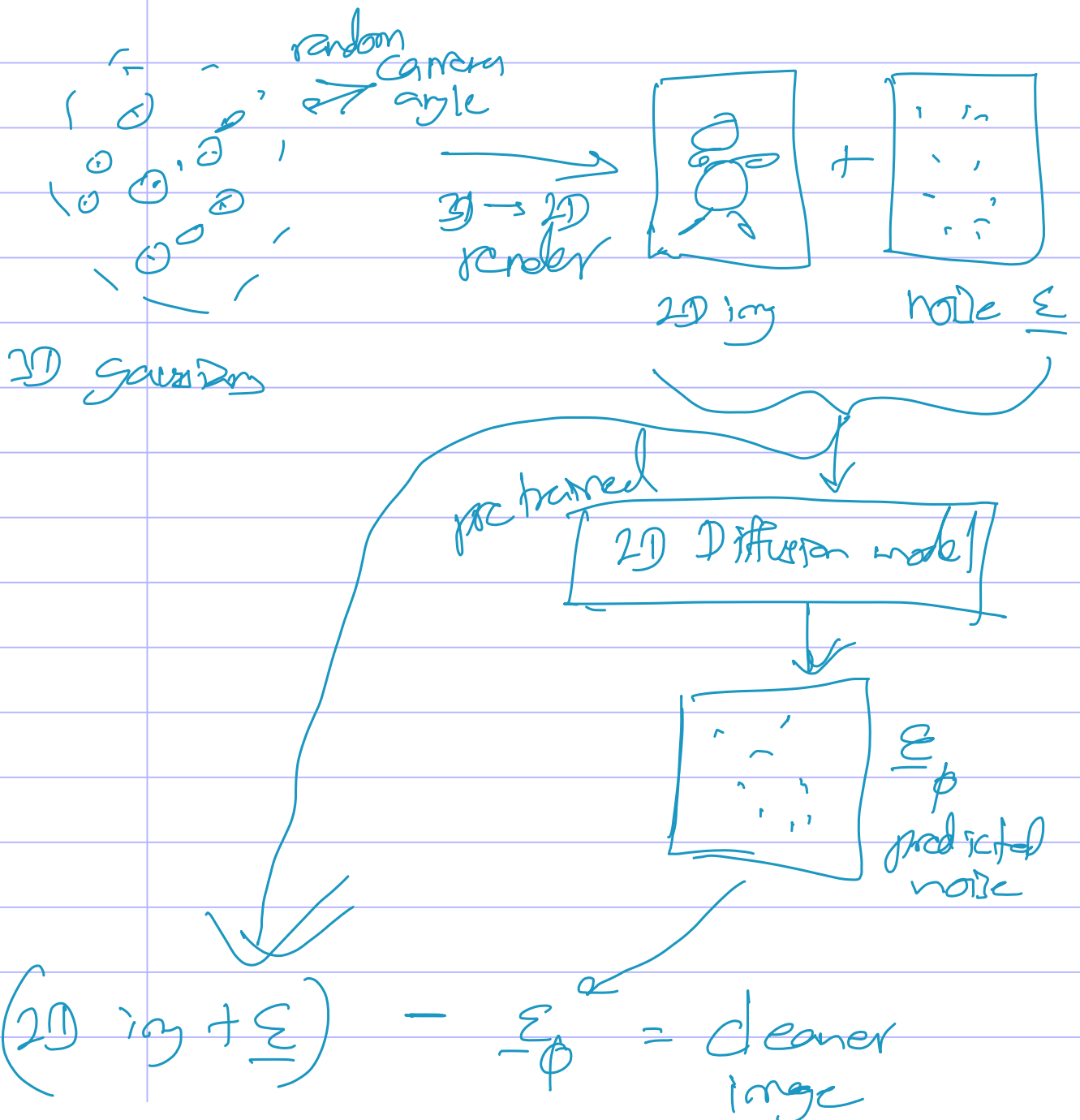


9840 Gen AI

1 Dream Generator

SDS - Score Distillation Sampling



or

$$2D \text{ img} - (\underline{\Sigma}_\phi - \underline{\Sigma}) = \text{clearer img}$$

$\Rightarrow \underline{\Sigma}_\phi - \underline{\Sigma}$ is the direction of optimality

* but we are working w 3D gaussian, not pixels.

* so we need to see how pixel values change w change in gaussian params

$$\Theta_i = \{ \underline{x}_i, \underline{\Sigma}_i, \underline{z}_i, \alpha_i, \underline{c}_i \}$$

where $\underline{x}_i$ = 3D position of center

$\underline{\Sigma}_i$ = scaling along 3 axes

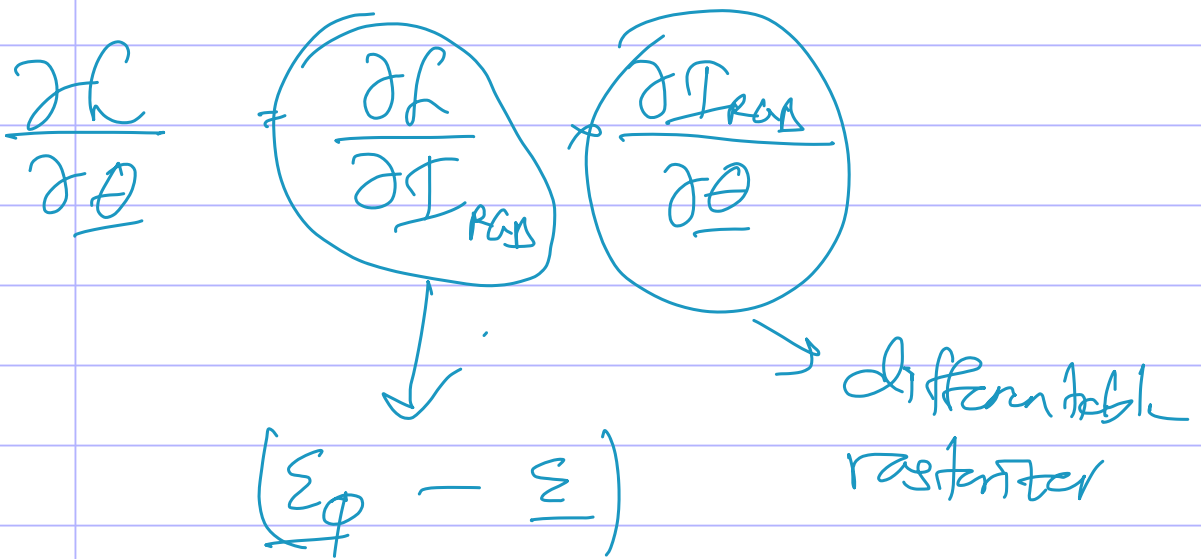
$\underline{z}_i$ = rotation quaternion

α_i = opacity

$\underline{c}_i$ = color. (no spherical harmonics in this version)

* $\frac{\partial I_{RGB}^p}{\partial \theta}$ = how pixels change when gaussian parameters are changed.

* $\mathcal{L} \rightarrow$ loss in pixel space



$$\nabla_{\theta} \mathcal{L}_{SDX} = \mathbb{E}_{t, p, \Sigma} \left[w(t) \left[\Sigma \phi(I_{RGB}^p, t, I_{RGB}^r, \Delta \phi) - \Sigma \right] \frac{\partial I_{RGB}^p}{\partial \theta} \right]$$

weighting function depends on time t

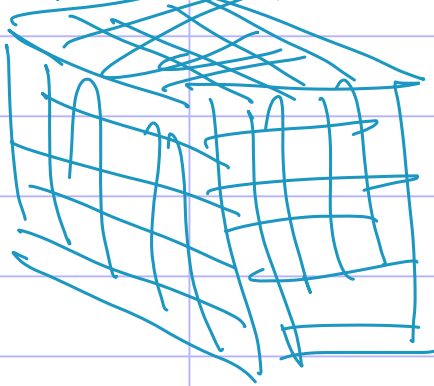
$\tilde{I}_{Ras}^r$: reference image given
as prompt to
the 2D $\rightarrow$ 1D task.

$$\mathcal{L}_{ref} = \lambda_{Ras} \| I_{Ras}^r - \tilde{I}_{Ras}^r \|_2^2 \\ + \lambda_A \| I_A^r - \tilde{I}_A^r \|_2^2$$

just to align the rendered
image & transparency at the
reference camera cycle to the
original input image.
(fidelity)

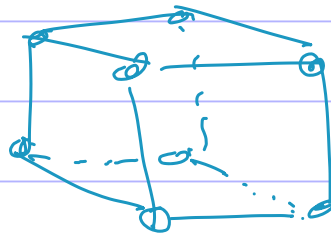
Mesh Extraction

Marching Cubes (circa '98)



← voxel grid.

Consider 1 voxel



← corners.

Refresher

selecting k from n options
(n choose k)

* Combination - don't care about order

$$\binom{n}{k} = \frac{n!}{k! (n-k)!}$$

$$\binom{8}{2} = \frac{8!}{2! \cdot 6!} = \frac{4 \times 7}{2} = 28$$

* permutation - care about order

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$P(8, 2) = \frac{8!}{6!} = 56$$

* # of unique configurations for 8 vertices each having 2 options.

$$2^8$$

* Marching cubes tries to identify if the surface of the object passes through a voxel cube.

* take 2 corners and see if it's inside or outside the object by comparing density.

* if all vertices have high density then the voxel is inside object

* if all have low density the voxel is outside.

* if some vertices have high density & some have low density, then the surface passes through the voxel. $\therefore$ this voxel is important for surface reconstruction

* in Dream-gaussian

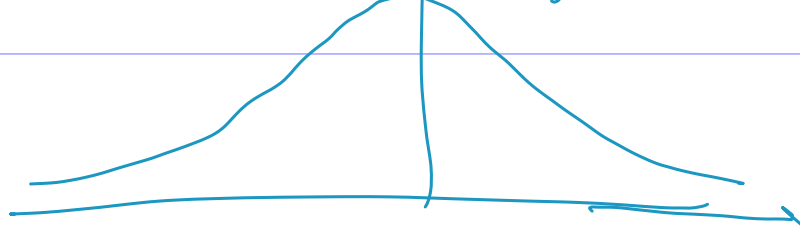
3D Gaussian $\rightarrow 128 \times 128 \times 128$ voxel grid.

density eqⁿ

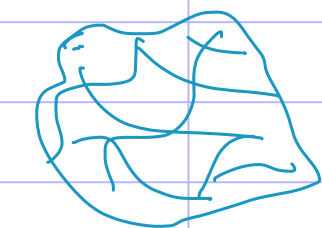
$$d(\underline{n}) = \sum_i \alpha_i \exp \left[-\frac{1}{2} (\underline{C}_{\underline{n} - \underline{u}_i})^T \Sigma^{-1} (\underline{C}_{\underline{n} - \underline{u}_i}) \right]$$

Sum of densities weighted by distance from center over all Gaussians intersecting at $\underline{n}$

weighted α_i



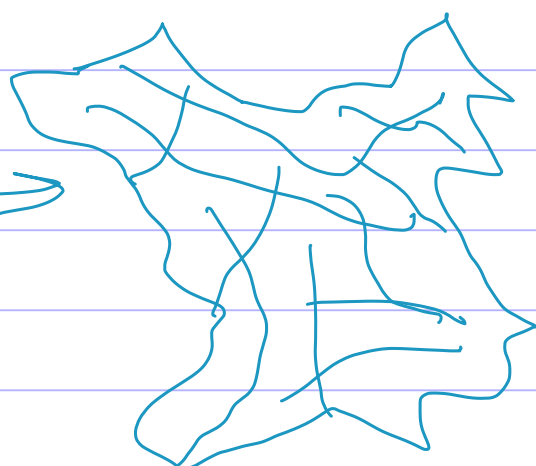
UV space texture refinement



3D mesh

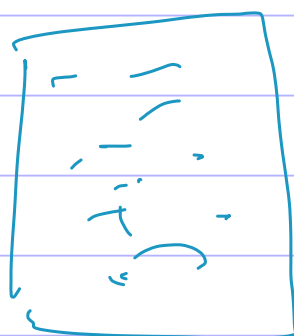


3D $\rightarrow$ 2D
flatten
mesh



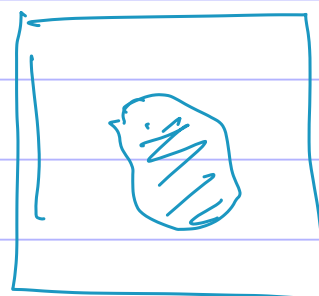
2D UV texture

render
image $\tilde{v}$
arbitrary angle ρ

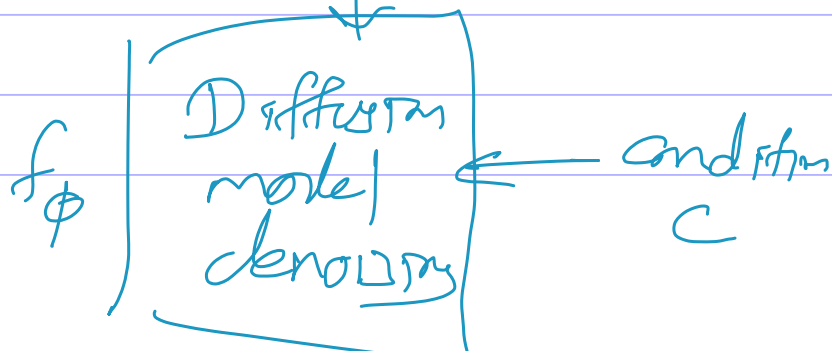


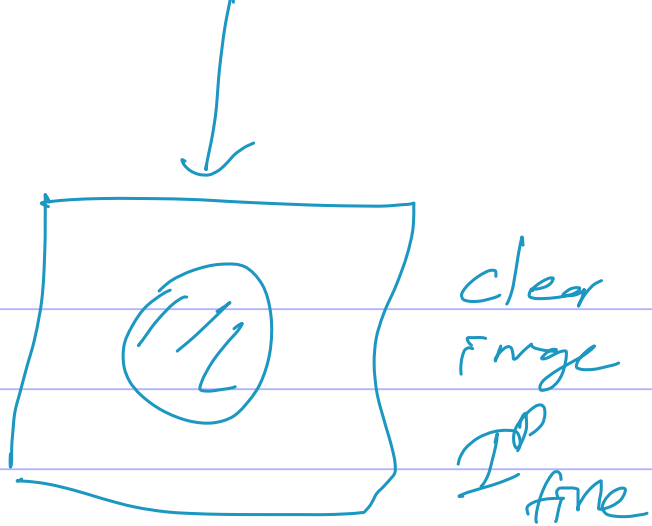
random noise
 $\Sigma(t_{\text{start}})$

+



I^p_{coarse}





$$I^p_{fine} = f_{\phi} [I^p_{case} + \varepsilon(t_{start}) | t_{start}]$$

↑ idea from SDEdit
(stochastic differential editing)

* Now optimize the UV space texture
w/ MSE loss

$$L_{MSE} = \left\| I^p_{fine} - I^p_{case} \right\|_2^2$$